

Q) Compute the time complexity using frequently Count for the following syntax.

① for ($i=1$; $i \leq 10$; $i=i+2$) — 1, 6, 5
 $\{$ $i = i \& 10$; — $5 \times 2 = 10$
 $\text{printf} ("i.d", i)$; — 5
 $\}$ — 27

② $i=1$;
 $\text{while} (i \leq 50)$;
 $\{ \text{printf} ("hello")$;
 $\}$

③ for ($i=1$; $i \leq 20$; $i++$) (1, 21, 20) = 42

④ for ($i=1$; $i \leq n$; $i=i+1$) ($1, n+1, n$)
 $\{ \text{printf} ("hello")$;
 $\text{printf} ("world")$;
 $\text{printf} ("i=%d", i)$;
 $\}$ — $5n+2$